

EMS501U – Cam Damper Masses for F1 Power Unit



Cara Fox

Intellectual Property Notice

- Due to the nature of F1, the following case study details have been modified slightly in terms of camshaft material, density limit and other select details to protect our competitive advantage.
- Please rest assured that there is still an answer to this problem! It's just our 2010 solution, not our 2025 one <3

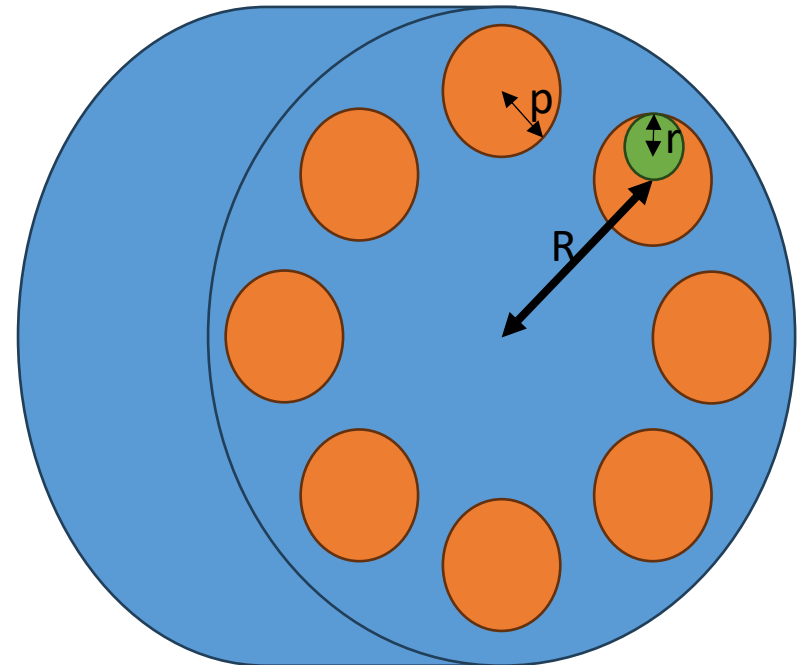
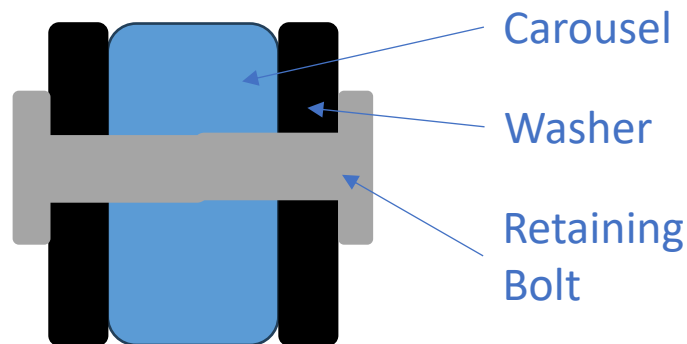


Cara Fox

Mechanical Engineering Graduate @
Mercedes AMG High Performance
Powertrains

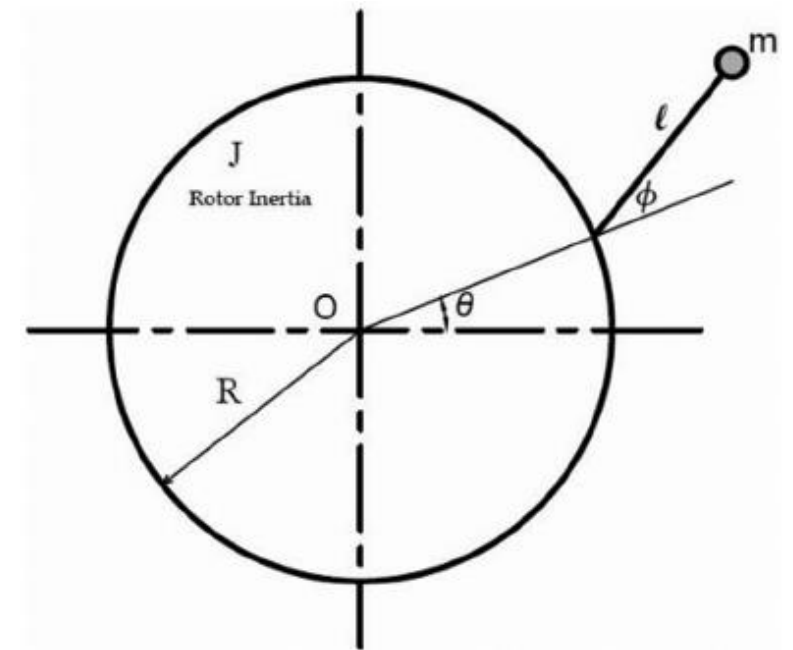
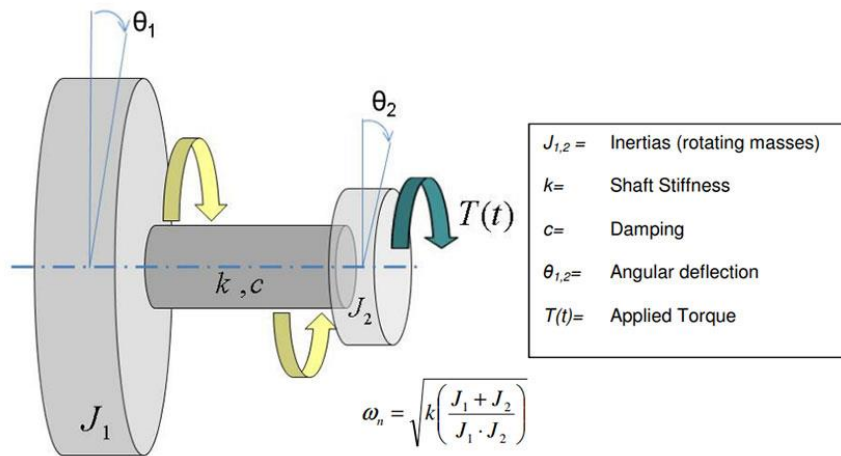
Rotational Dampers – how it works

- Rotational dampers oppose vibrations in a torsional system to reduce the fatigue on components. The less the component moves back and forth the longer it will last and the smoother the movement in the power unit will be.
- [Pendulum Dampers](#) oppose this movement with a high-density weight that swing in the opposite direction to motion. E.g. as the carousel below (blue) rotates clockwise the damper mass (green) will rotate anti-clockwise in the bore (orange).
- The mass is held in the bores by two washers on either end. The washers and carousel are made of the same material. So, when considering wear, you should consider your chosen material interacting with EN9 steel. Consider both yield strength and hardness.



Brief

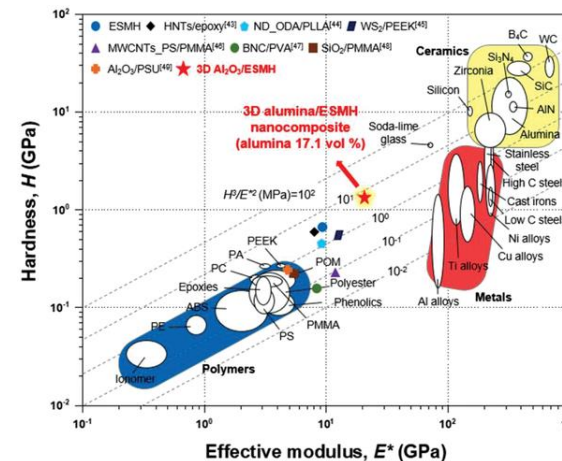
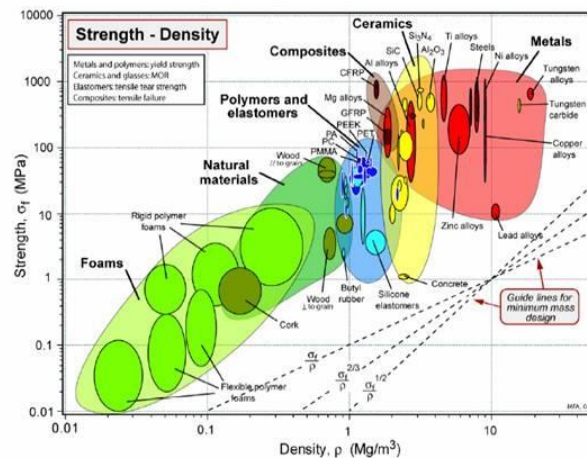
Coming into the 2010 season, some materials have been made illegal by FIA regulations. This includes our current material for the camshaft pendulum dampers. The material must now be below a **density of 18,100 kg/m³**. Packaging of the carousel is restricted to 18mm diametrically and you may fit as many pendulum dampers as you like per carousel. The aim of this project is to find the solution that will produce the **least wear on the damper masses or carousel** while **matching the damping torque and vibration order**. The camshaft carousel and washer to assemble the system are made of **EN9 steel**.



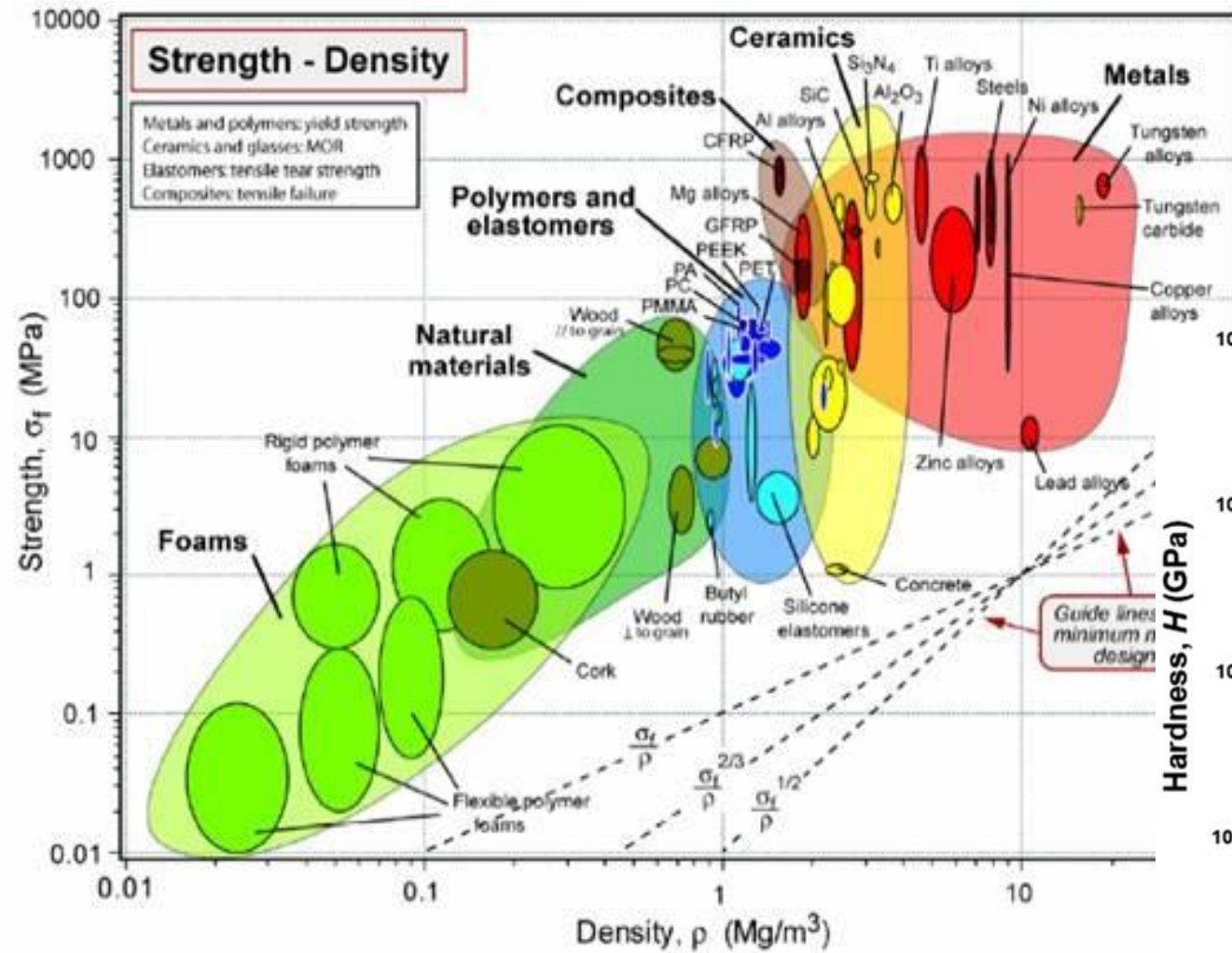
Brief

- Less than a density of 18,100 kg/m³.
- Least wear on the damper masses or carousel
- Match the damping torque and vibration order.
- Rolling face interaction with EN9 steel. (Hardness = 150-350HV)
- Operating in a power unit – exposed to temperatures up to 140 degC, oil, vibration within the carousel

Constraints	Objectives	Free Variables
Density < 18,100 kg/m ³ Material types i.e. must be a solid Hardness around EN9 steel (100 < HV < 400) Max Temperature > 140 degC Corrosion resistant to oil	Wear Resistance – high ((Hardness * Contact Area) / (Sliding Velocity * Load)) Hardness – high Density – high Yield Strength – high	Cost

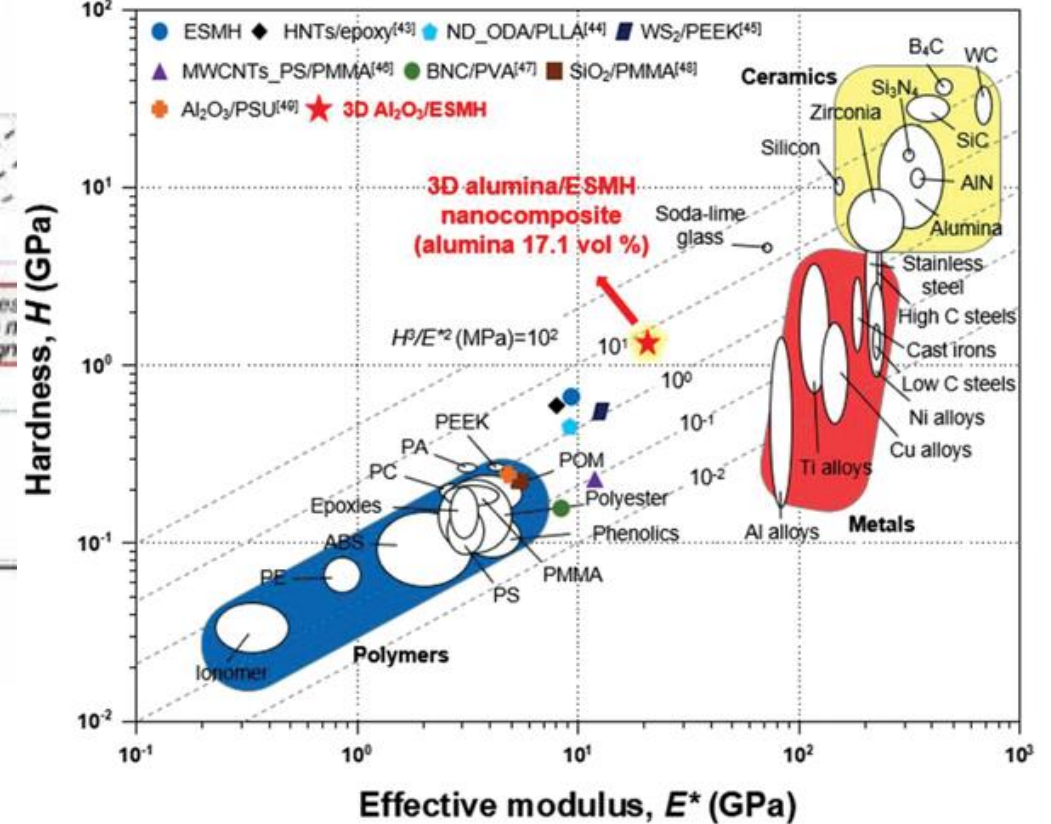


Brief



Options:

Tungsten Alloys
Tungsten Carbide
Nickel Alloys
Steel

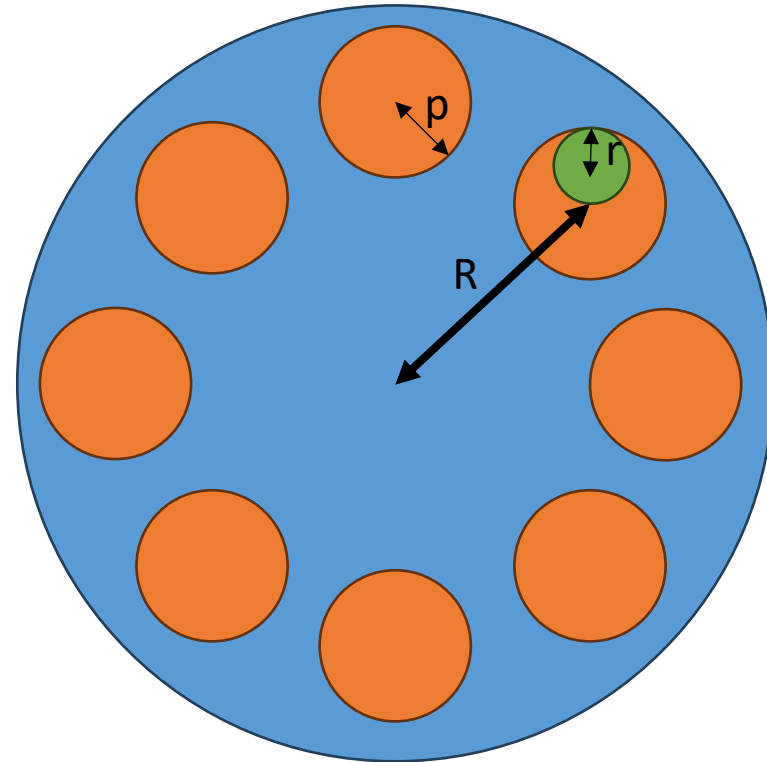


Brief

Material	Density (kg/m3)	Hardness (HV)	Yield Strength (MPa)	Cost (£ per kg)
Tungsten Alloys	17000-18500	300-400	500-1000	75-230
Tungsten Carbide	14500	1600-2500	300-350	200-300
Nickel Alloys	8000-9000	100-400	150-170	40-75
Steel	7000-8000	100-800	200-500	100-200

Calculations

- Damper Mass is a simple cylinder
 - Radius = designable
 - Length = 15mm
- Damper Torque:
 - $$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$
 - **Your damper torque should equal as close to 19.67Nm @ 11000rpm**
- Vibration Order:
 - $$\left(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))} \right)^{1/2}$$
 - **The vibration order of this system is 4**



m = mass of damper

ω_{engine} = engine speed

R = carousel radius

$e = p - r$

r = roller radius

p = bore radius

$n_{dampers}$ = number of dampers (and bores)

Calculations

$$\frac{m \times \omega_{engine} \times R \times 0.5 \times (R + e - \frac{25}{r})}{1 + (\frac{25}{r})^2} \times n_{dampers}$$

$$(\frac{R}{e \times (1 + (\frac{37.5}{r^2}))})^{1/2}$$

Variable	Damper Torque	Vibration Order
Mass – higher	Higher	-
Bore Radius – higher	Higher	Higher
Roller Radius – higher	Lower	Lower (order of 2)
Number of dampers – Higher	Higher	-

Countering Wear in Practice

- **Material Selection**

- Use materials with higher wear resistance for the rolling/sliding components.
- Applying coatings such as chrome plating or specialized wear-resistant coatings can help reduce friction and protect surfaces from wear. For example, heat treatments, anodising or PVD/CVD (DLC - Diamond-Like Carbon) coatings can improve durability.

- **Lubrication**

- Ensure that proper lubrication is applied to the moving components. This helps reduce wear by minimizing direct contact between moving parts.

- **Friction Control**

- Ensure that the damper is designed so there is enough clearance between components. Surface finish of materials will make a huge difference, therefore machining methods should be considered for both carousel and damper mass.

Areas of Success

Identifying material properties

Verbal communication

Areas for improvement

Tolerancing and packaging

Damper Mass = Dense

Cost

Visual Presentation

Meeting the CORRECT requirements